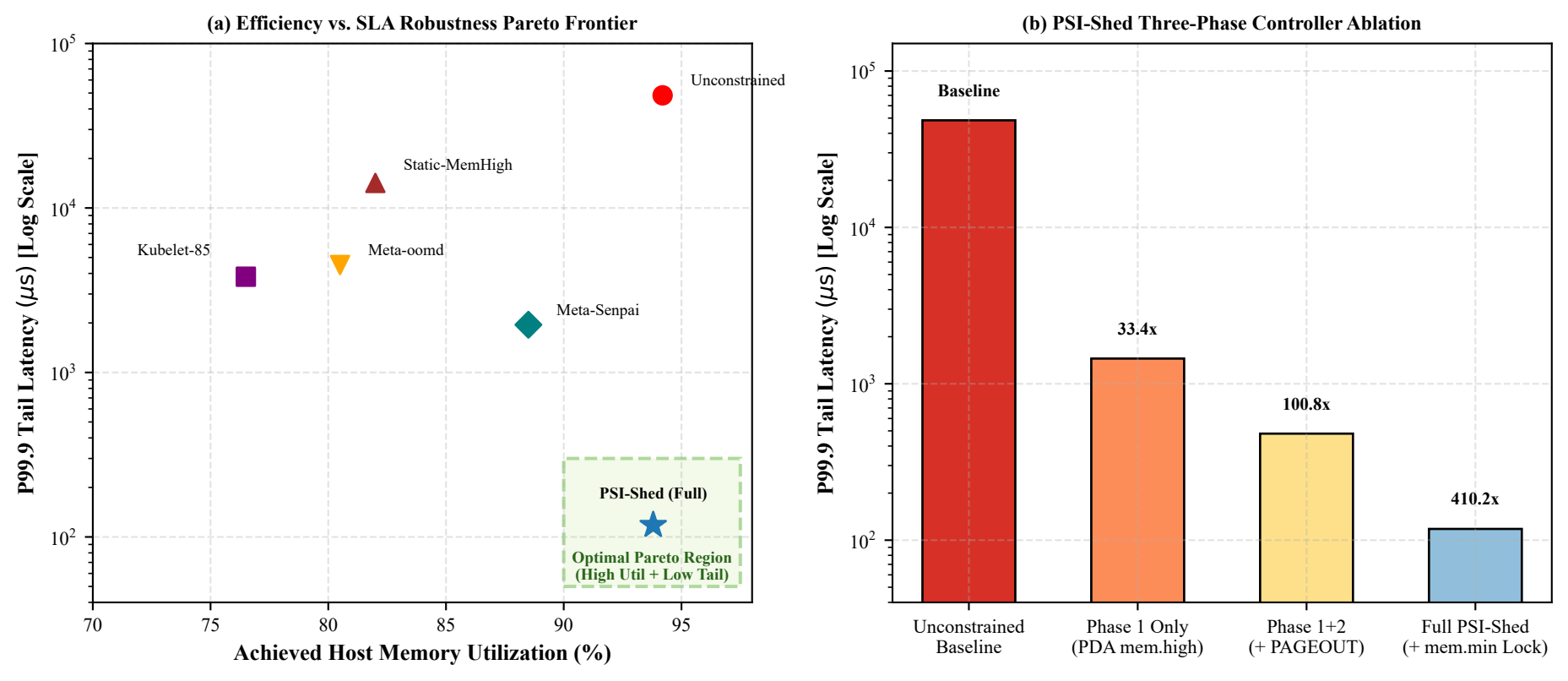

(a) Efficiency vs. SLA Robustness Pareto Frontier
P99.9 Tail Latency (μs) [Log Scale]
Achieved Host Memory Utilization (%)
Unconstrained
Static-MemHigh
Kubelet-85
Meta-oomd
Meta-Senpai
PSI-Shed (Full)
Optimal Pareto Region
(High Util + Low Tail)
(b) PSI-Shed Three-Phase Controller Ablation
P99.9 Tail Latency (μs) [Log Scale]
Baseline
33.4x
100.8x
410.2x
Unconstrained Baseline
Phase 1 Only (PDA mem.high)
Phase 1+2 (+ PAGEOUT)
Full PSI-Shed (+ mem.min Lock)